



COLLEGE OF ENGINEERING, CONSTRUCTION, AND LIVING SCIENCES

BACHELOR OF INFORMATION TECHNOLOGY (BIT)

CHECKPOINT-2, SEMESTER 2, 2026

ID616001 OPERATING SYSTEMS CONCEPTS

LEVEL 6, CREDITS 15

VERSION 1.1

General Instructions

This checkpoint contributes 10% to your total course grade. It is focused entirely on **Operating System process management** with **Bash scripting**.

- This checkpoint has a **task** worth 100 marks. You must attempt every sub-task.
- You can consult your linux help/man page, slides, previous lab solutions, source codes, and class notes.
- All work must be completed independently. **Collaboration is strictly prohibited.**
- All solutions must be your own original work.

Submission Instructions

- Complete the task using the **virtual machine (VM)** assigned to you on the Azure cloud platform. **Do not use any other VM.**
- Document every step of your solution. Your documentation must include:
 - Screenshots of commands and outputs.
 - Any scripts or configuration files used.
 - Descriptive captions explaining what each screenshot or script shows.
- Prepare a report containing your solution, evidence, and explanations. Ensure that answers and supporting material are placed in the designated response area for each sub-task.
- Submit the following:
 - Your report saved as: `<username>_CheckPoint2_S22026.docx`.
 - All supporting files (scripts, screenshots, configuration files, etc.).
- Package all required files into a single **.zip** archive and upload it to the **Assessment** section of the Microsoft Teams course channel.

Learning Outcomes

- Analyse and manage process, memory, storage, and network management modules in operating systems.

System Resource Monitor with Process Control (100 Marks)

Scenario

Your role as a junior system administrator involves both **monitoring** the system and **controlling** processes that are misbehaving. A runaway process consuming 90% CPU might just need its priority lowered; a frozen process might need a polite termination request before being force-killed.

You will write a single Bash script that first reports the busiest processes on the system, then offers an interactive menu of actions the operator can take on any chosen process: **terminate gracefully** or **change priority**.

Your Task

Write a script named `system_monitor.sh`. It must implement all of the following sub-tasks.

Part A - CPU/Process Monitoring

1. Extract the top 5 CPU-consuming processes (5 marks).
2. Display the top processes with their PID, CPU usage, memory usage, and command in the following format (10 marks):

```
PID: 2345 | CPU: 70.0% | MEM: 35.2% | CMD: /usr/sbin/mysqld
```
3. Calculate and display the sum of CPU usage for these top processes. (10 marks).
 - **Hint:** To calculate the total CPU usage, consider using a loop or `awk` to sum the third column (%CPU) of your top process list. You can format the output using `printf` for one decimal place.
 - **Example: Summing a Column Using awk** Suppose `numbers.txt` contains:

```
10
20
30
40
```


You can sum the numbers with:

```
total=$(awk '{sum += $1} END {printf "%.1f", sum}' numbers.txt)
echo "Total: $total"
```
4. Prompt the user: `Terminate them all? (y/N)`. If confirmed, terminate the processes using `kill -9`, otherwise exit gracefully. (5 marks)
5. If no high-CPU processes are found, display: `No processes found to monitor.` and exit. (5 marks)

Part B - Memory Monitoring

6. List the top 5 processes by memory usage in this format (10 marks):

```
PID: 2050 | COMMAND: nginx | MEM: 15.0%
```
7. Check total system memory usage. (10 marks)
If it exceeds 80%, append a timestamped warning to `/var/log/ops_monitor.log`, for example:

```
[YYYY-MM-DD HH:MM:SS] WARNING: Memory usage at 85%
```

Key Commands & Hints (Parts A and B)

- Use `awk` or `cut` to parse `ps` output.
- Use `bc` or Bash arithmetic to calculate total CPU usage.
- `date '+[%Y-%m-%d %H:%M:%S]'` for consistent timestamps in logs.

Part C - Interactive Process Control

After the monitoring report is printed, the script should present an interactive prompt that lets the operator act on *any* PID from the system. You do not need to restrict actions to the 5 processes you displayed — the operator may type any valid PID.

8. **Prompt for PID and action.** (5 marks)
Ask the user for a PID and a desired action. Suggested format:

```
Enter PID to act on (or press Enter to exit): 4321
Choose an action -- [t]erminate, [n]renice: t
```


If the user just presses Enter at the PID prompt, exit cleanly.

9. Process-alive check. (5 marks)

Before applying any action, check whether the chosen PID is still alive (use `kill -0 <pid>`). If it isn't, print `PID <pid> is not running.` and skip the action.

10. Terminate: SIGTERM, then escalate to SIGKILL. (15 marks)

When the user chooses `terminate`:

- Send SIGTERM to the PID. Print: `Sent SIGTERM to <pid>.` Waiting 3 seconds...
- Wait 3 seconds (`sleep 3`).
- If the process has exited, print: `Process <pid> stopped gracefully.`
- If it is still alive, escalate to SIGKILL (`kill -9`) and print: `Process <pid> did not stop -- sent SIGKILL.`

11. Change priority live (renice). (20 marks)

When the user chooses `renice`, prompt for a new nice value:

Enter new nice value (0..19): 10

- Validate the input: must be an integer in 0..19. Reject any other input with: `Error: nice value must be in 0..19.` and exit the action.
- Apply with `renice -n <value> -p <pid>`.
- Confirm with: `Process <pid> set to nice value <value>.`

Key Commands & Hints (Part C)

- Reading user input:

```
read -rp "Enter PID to act on (or press Enter to exit): " pid
```

- Action dispatch with case:

```
case "$action" in
    t|terminate) ... ;;
    n|renice)    ... ;;
    *)          echo "Unknown action" ;;
esac
```

- Check a process is alive without killing it: `kill -0 <pid>` (exit status 0 = alive, non-zero = dead).

Course Regulations

For comprehensive information regarding course regulations, please refer to the course directive. This includes details on:

- **Late Submission Penalties:** Understand the consequences and penalties associated with submitting assignments after the deadline.
- **Referencing Style:** Follow the required citation and referencing format for all sources used in your work.
- **Plagiarism Policy:** Review the guidelines on academic integrity and the consequences of plagiarism.
- **Additional Information:** Any other relevant regulations and procedures that pertain to assessments.

Ensure that you are familiar with and adhere to these regulations to maintain academic integrity and meet course requirements.